Kumar Manthan

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EDUCATION

Vellore Institute of Technology, Chennai *2023 – 2027*

B.Tech, Computer Science and Engineering (AI & ML) | CGPA: 8.37/10

TECHNICAL SKILLS

Programming: C++ (Primary), C, Java, JavaScript, Python

Frontend: HTML, CSS, Vanilla JavaScript

Backend: Node.js, Express.js, REST APIs

Databases: PostgreSQL (Primary), MongoDB

Authentication & Security: JWT, bcrypt, OAuth 2.0, SHA-256, Ed25519

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

PROJECTS

Posturector

Personalized Adaptive Lumbar Posture Monitoring & Correction System

React Native, Expo, TypeScript, Node.js, Express.js, SQLite, IMU Sensors, REST APIs

- Developed a real-time lumbar posture monitoring system using IMU sensors, accelerometer–gyroscope sensor fusion, and complementary filtering.
- Implemented personalized calibration, baseline-relative deviation detection, stability and persistence filtering, and real-time haptic feedback to reduce false corrections.
- Designed a behavioral adaptive threshold controller using five longitudinal posture features and multi-timescale EMA-based evidence accumulation.
- Integrated Borg RPE and EMG-based fatigue measurements to learn data-driven weights for five behavioral features and improve personalized adaptive threshold decisions.
- Implemented bounded, reversible threshold adaptation with severe-deviation and instability safety gates, and adaptive feedback based on correction-response history.

MediChain

Secure Medical Record Integrity System

Python, Flask, SQLite, JavaScript, HTML, CSS, Cryptography, REST APIs

- Developed an offline, lightweight medical record integrity system using SHA-256 hashing and Ed25519 digital signatures to verify record authenticity and detect unauthorized modifications.
- Implemented nonce-based challenge-response authentication and password-encrypted private-key storage for staff credentials.
- Designed hash-linked record versioning and tamper-evident audit logs to track updates and verify stored history.
- Built REST APIs and a web interface for staff registration, record creation, modification, version history, and integrity verification.

Touroholic

Full-Stack Tour Discovery, Booking & Review Management Platform

Node.js, Express.js, MongoDB, Mongoose, JavaScript, Leaflet.js, REST APIs, JWT

- Developed a full-stack tour discovery and booking platform with responsive tour listings, detailed itineraries, user profiles, personalized booking history, and a review and rating system.
- Designed the application using MVC architecture and built RESTful APIs for managing tours, users, bookings, reviews, and authentication.
- Implemented JWT-based authentication with registration, password hashing and recovery, profile management, and role-based access control.
- Integrated interactive Leaflet and OpenStreetMap maps to visualize tour locations and itinerary stops.

COMPETITIVE PROGRAMMING

LeetCode: 500+ DSA problems solved; Knight (Max Rating: 1866); 50+ SQL problems solved

CodeChef: 100+ DSA problems solved; 2-Star Coder (Max Rating: 1454)

GeeksforGeeks: 150+ DSA problems solved